

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 8: All-Weather Hazards: Windshear, Ash & Wake Turbulence

Syllabus Source: CAE Oxford (pp. 273-310) | Pass Target: >= 90% AviationExam

1. WINDSHEAR RECOGNITION & ESCAPE MANEUVER

Low-level windshear produces sudden changes in indicated airspeed (> 15 kt), vertical speed (> 500 ft/min), and pitch angle (> 5°):

- Immediate Escape Maneuver: 1. Disconnect autopilot; 2. Advance thrust levers to FULL TOGA (firewall power!); 3. Rotate wings-level towards the pitch limit indicator (stick shaker); 4. STRICTLY PROHIBITED: DO NOT CHANGE GEAR OR FLAP CONFIGURATION until terrain clearance is assured (changing gear causes massive temporary drag increase!).

2. ICAO WAKE TURBULENCE SEPARATION MINIMA

Wake vortices trail from wingtips, descending at 400 to 500 ft/min. Maximum hazard occurs behind Heavy aircraft in clean, slow, high-angle-of-attack flight:

Leading Aircraft Category	Following Aircraft Category	Departure Time Separation (Same Runway / Intersection)
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HEAVY (>= 136,000 kg)	MEDIUM (7,000 to 136,000 kg)	2 MINUTES (3 minutes if following from intermediate intersection).
HEAVY (>= 136,000 kg)	LIGHT (< 7,000 kg)	2 MINUTES (3 minutes if following from intermediate intersection).
SUPER (Airbus A380)	HEAVY / MEDIUM	3 MINUTES full runway (4 minutes from intersection).
SUPER (Airbus A380)	LIGHT	4 MINUTES (5 minutes from intersection).

3. VOLCANIC ASH CLOUD ENCOUNTER PROCEDURES

- Engine Flameout Threat: Volcanic ash melts at 1,100°C inside turbine combustors, fusing into glassy enamel that suffocates turbine cooling holes and causes ALL ENGINES TO FLAMEOUT!
- Escape Actions: 1. Execute IMMEDIATE 180° TURN back out of ash cloud; 2. Retard thrust levers to IDLE (reduces combustion temperature below ash melting point); 3. Start APU; 4. Turn on continuous engine ignition; 5. Select maximum cabin air bleed.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Wake Turbulence Avoidance Rule

Takeoff Behind a Heavy Jet: - Lift-off occurs at rotation (V_r). Lift-off BEFORE the preceding aircraft rotation point and climb above its flight path. - Minimum ATC separation: 2 minutes (3 minutes from intermediate intersection). Landing Behind a Heavy Jet: - Stay ABOVE the preceding aircraft glidepath. - Touch down BEYOND the preceding nose wheel touchdown point (vortices cease upon nose wheel contact).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: During an in-flight encounter with low-level WINDSHEAR, why is changing flap or landing gear configuration

- [A]: Flap motor will overheat
- [B]: Retracting or extending gear/flaps creates significant TRANSIENT DRAG and loss of lift, drastically increasing the risk of ground impact
- [C]: ATC bans gear changes
- [D]: Altimeter will fail

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The transient drag during gear door cycling can cause immediate loss of the precious few feet of altitude remaining.

Q2: What is the mandatory ICAO wake turbulence departure time interval between a HEAVY aircraft taking off a

- [A]: 1 minute
- [B]: AT LEAST 2 MINUTES
- [C]: 5 minutes
- [D]: 30 seconds

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. ICAO Doc 4444 requires a minimum 2-minute separation behind a Heavy aircraft for same-runway departures (increased to 3 minutes for intersection take-offs).

Q3: When inadvertently penetrating a volcanic ash cloud, why should engine thrust levers be retarded to IDLE

- [A]: To save fuel
- [B]: To LOWER TURBINE GAS TEMPERATURES below the melting point of volcanic ash (~1,100°C), preventing molten glass from accumulating on turbine blades
- [C]: To increase airspeed
- [D]: To pressurize cabin

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At idle thrust, EGT drops below the ~1,100°C melting point of silicate ash, stopping the buildup of molten glassy deposits on turbine guide vanes.